

Integrating Weather Patterns into Machine Learning Models for Improved Electricity Demand Forecasting in Sri Lanka

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Abstract— The electricity demand in Sri Lanka is expected to increase steadily over time. Planning for future demand and ensuring an adequate electricity supply poses a significant challenge. It is crucial to accurately forecast future demand in order to maintain an uninterrupted power supply. Previous studies have explored the correlation between weather factors and electricity demand with the aim of accurately predicting demand values. Thus, the objective of this study is to forecast the monthly electricity demand in Sri Lanka, by considering the influence of weather patterns. In this study, rainfall, humidity, and temperature weather parameters, along with historical monthly demand data, are taken into consideration. The identification of the most crucial weather variables is based on their correlation with electricity demand data. Various techniques have been employed for forecasting electricity demand over the past decade. However, the limitation of previous studies lies in their failure to incorporate past weather data alongside electricity demand data. This gap is addressed in the present study. This study used Vector Auto Regression (VAR) and Long Short-Term Memory (LSTM) models to forecast monthly electricity demand in each district of Sri Lanka. The VAR model demonstrated lower values by comparing the performance metrics, including Root Mean Square Error, Mean Square Error, Mean Absolute Error, and Mean Absolute Percentage Error. As a result, the VAR model was chosen as the most suitable model for forecasting monthly electricity demand by incorporating weather variables.

Keywords— *electricity demand, weather patterns, forecasting*

I. INTRODUCTION

Sri Lanka is a tropical island in the Indian Ocean. Sri Lankans face different weather patterns throughout the year. Some months are really hot, but some months are cool with heavy rainfall. The rainfall pattern is influenced by the Indian Ocean monsoon winds. Sri Lanka has two monsoons, the Northeast Monsoon (December to March), and the Southwest Monsoon (June to October). Humidity also depends on seasonal patterns of rainfall and region. In this way, frequent fluctuations in weather patterns can be seen in Sri Lanka. Hydropower already plays a major role in Sri Lanka's electricity sector. Also, solar and wind energy can be the main energy generators for Sri Lanka as a tropical island. Therefore, weather patterns have a major impact on Sri Lanka's electricity supply. Also, electricity demand also changes according to such volatile weather patterns. There are many articles that have proven how Sri Lanka's electricity demand will grow rapidly in the future. So far, little attention has been paid to forecasting the country's electricity demand [1]. On the other hand, there should be a national plan to increase the use of renewable energy sources for electricity production in the country to achieve sustainable development goals. The

continuously rising electricity demand is evident due to trends in domestic and industrial consumption in Sri Lanka [2]. During the last fifteen years, electricity demand in the country has been growing at an average rate of about 4.4% per annum while peak demand has been growing at a rate of 2.6% per annum [3]. With the end of the almost 30-year civil war in 2009, the country experienced a notable amount of human and social development, which has increased electricity demand in the residential, commercial, and industrial sectors. The highest recorded electricity demand in Sri Lanka during 2016 was 2406.4 MW, an 8.9% increase over the demand in 2015. The demand for electricity is expected to increase by roughly 6.5% annually on average. This study detailed how, following the Sri Lankan civil war, the country's electricity demand climbed year after year [4]. On the other hand, Sri Lanka's primary source of energy generation is now thermal sources, which can account for up to 80% of the supply during severe droughts [5]. This idea has shown how important it is to reduce energy wastage. For that, the electricity demand should be known correctly. Modeling electricity consumption with high accuracy is essential for the effective development of energy policies for good demand forecasting and to minimize costly errors. The ability to predict electricity demand has also been helpful to customers [6]. All those studies provide adequate evidence as to why electricity demand is the most critical predictor for achieving a sustainable supply-demand balance. This research attempts to provide an accurate electricity demand forecasting mechanism to fill that gap.

A System Control Centre (SCC), is the center of operational activities of the CEB. Short-term demand forecasting is one of the primary prerequisites for operations planning. For SCC, there is still no sophisticated forecasting system. The CEB currently does not forecast regional demand, despite the fact that it is a need. [1]. The use of forecasting methodologies including Multiple Regression (MRA), Exponential Smoothing, Iterative Reviewed Least Squares, Stochastic Time Series, Fuzzy Logic, Artificial Neural Networks (ANN), and Expert Systems has advanced significantly. Such techniques for forecasting electricity demand, however, have not yet been used in Sri Lanka. In Sri Lanka, system engineers used historical electricity usage data and their own judgment to forecast the following day's demand [7]. The above articles have strongly justified the need for an accurate electricity demand forecasting system for Sri Lanka. Furthermore, the current long-term electricity demand forecasting studies of the Ceylon Electricity Board are conducted by considering possible explanatory variables with an econometric approach based on multiple linear regression. It has also expressed the importance of developing an AI-based non-linear forecasting approach with explanatory

variables, including climate conditions, that will justify and compare the performance of official forecasts. Therefore, this research has shown that it is essential to consider the relationship between electricity demand and weather variables in order to obtain an accurate forecast of electricity demand [8]. Therefore, the purpose of this study is to forecast the electricity demand in Sri Lanka, taking into account the influence of weather patterns. According to the literature review, many factors have been found that affect the determination of electricity demand. But among them, weather parameters are the main factor for electricity demand forecast. So, here the electricity demand in Sri Lanka is to be forecasted using weather patterns.

II. LITERATURE REVIEW

A. Different Types of Demand Forecasting Approaches

Many studies have been done on electricity forecasting approaches. There are two main groups of widely used energy demand and generation forecasting techniques. Statistical or mathematical methods fall under the first group, whereas modern statistical learning-based methods go under the second (also known as machine learning)[9]. All the decomposition approaches in forecasting the electricity demand by identifying the lack of comprehensive review concerning the feature of the decomposition approaches employed in forecasting the electricity demand. Medium-term load forecast (monthly, weekly) is influenced by meteorological factors, and economic development factors. Also commonly used methods are the neural network, regression models, Support Vector Machine (SVM), and time series models for electricity demand forecasting [10]. Articles on electricity forecasting published from 2016 to 2019 have been collected based on a systematic literature review method to find the most accurate machine learning methods for short-term electricity forecasting. The conclusion has been made that the dataset size may have an impact on the accuracy of the model created and that different researchers have chosen different machine-learning algorithms and metrics for the evaluation. The most widely used machine learning algorithms among them are linear regression, support vector regression (SVR), ANN, and SVM. Several articles have used input variables like calendar and energy usage data [11]. Optimized Multivariate adaptive regression spline(MARS), SVR, and Autoregressive Integrated Moving Average (ARIMA) models have been developed for electricity load forecasting in Queensland, Australia. The MARS model is best for time horizons of 0.5 h and 1.0 h. The SVR model is best for 24 hours' time horizons. They did not use any weather variables in developing the model. But, in future studies, they hope to use electricity demand data linked with weather factors for accurate forecasting [12]. Based on Support Vector Machine for Regression (SVMR), accurate and efficient energy forecasting tools have been provided. Four commercial buildings are randomly chosen for the case study, and the models are created using monthly utility costs from the landlord. Careful analysis of available data reveals the most influential parameters are humidity, average temperature, and solar transmissions to forecast electricity demand in Colombo, Sri Lanka commercial buildings. They have only used Support Vector Machine Regression to model commercial buildings' energy demand. They did not consider other models. Also, the sample data set is very limited and only considers the electricity demand of 4 buildings [13]. The study developed an ANN based on socio-economic indicators and climatic

conditions for long-term annual electricity demand forecasting in Sri Lanka. After observing the results, the ANN model has shown high accuracy for past validation data (0.03% error compared to true). But in that study, they used only the ANN model for electricity demand forecasting [8]. An artificial neural network and a regression model were developed to forecast long-term electricity consumption in Thailand. Gross Domestic Product (GDP), population, maximum temperature, and electrical power demand are used as inputs to a neural network to predict electrical consumption. The outcomes demonstrate that the ANN model can deliver a result that is more accurate than the regression approach[14]. ANN algorithms Levenberg Marquardt Back Propagation, Bayesian Regularization, and Scaled Conjugate Gradient algorithms were developed for short-term load forecasting. The data used in the forecast are hourly historical data of temperature and electricity demand in Toronto, Canada. No weather factor other than the temperature variable is considered as input. According to the conclusion, Levenberg Marquardt Back Propagation and Bayesian Regularization algorithms were good for prediction, but the Scaled Conjugate Gradient algorithm was not good for short-term forecasting [15]. The study was conducted to implement artificial neural networks to model and forecast half-hourly electricity load demand in Tunisia during 2000-2008. The Levenberg-Marquardt training algorithm performs well in predicting half-hourly electricity demand, according to experimental results. The temperature is taken into account as a meteorological variable in that research as well only[16]. A model of Long Short-Term Memory (LSTM) artificial neural networks was created to forecast Poland's demand for electricity. For a small set of statistical data models, LSTM, a deep learning technique, represents long-term relationships in time-series data. The outcomes demonstrate that the suggested model can greatly increase prediction accuracy [17]. This paper proposed a demand forecasting method based on multi-layer LSTM networks. The proposed method is compared with well-known time series forecasting methods such as ARIMA, ANN, K-nearest neighbors(KNN), Recurrent Neural Networks(RNN), SVM, and single-layer LSTM. Experimental results have shown that multi-layer LSTM is superior in demand forecasting [18]. In this study, a Vector Auto Regression (VAR) model was used to forecast three crucial weather variables for 61 American cities. The three variables were modeled as response variables across all locations. The link between observations in adjacent eras has been described using lag words. Based on the results, the proposed VAR time series approach has been selected as appropriate for short-term forecasting. Mean absolute error, root means square error, and relative mean square error has been used to evaluate the performance of the VAR model [19]. The study's objectives were to create multivariate forecasting models and analyze temporal correlations between demands on emergency department (ED) and inpatient hospital key resources. Their results suggested that multivariate VAR models can be used for reliable forecasting [20]. The study was done to forecast the energy consumption of three distribution networks in the city of Tetouan, Morocco, using linear regression, decision tree, random forest, feedforward neural network, and support vector machine for regression. The results show that the random forest model has outperformed other models in predicting electricity consumption in Tetouan city. This study used a limited data set and was limited to only 3 distribution networks in Tetouan city [21]. The study was done for predicting energy

consumption in a given period of time based on the performance of two models, ARIMA and NAR. The performance of the NAR model is compared with the ARIMA model, which is widely used for time series analysis. It has been shown that both the ARIMA model and Nonlinear Autoregressive Neural Networks(NAR) model can obtain good forecasts and can be used effectively for energy forecasting. Although the performance of the ARIMA model is better than NAR model, it is observed by comparing the prediction error values. In order to reduce the complexity, they have reduced the inputs used to generate the model. In this paper, more simple methods are proposed that use only available data on energy consumption. So, they did not specify any weather variables as input parameters [22]. The ARIMA model and SARIMA model are applied separately to forecast the load data. It is concluded that the Seasonal Autoregressive Integrated Moving Average(SARIMA) model gives better results considering the seasonality of load data. In their study, they used only daily electricity load data from Assam, India. They did not use variables other than load data as input parameters to develop the model. They did not use any machine learning methods other than 2 traditional statistical methods to forecast electricity demand [23]. Based on previous studies and review findings, it has been suggested mostly to use ANN, SVR, VAR, LSTM, or time series (SARIMA) models to accurately forecast electricity demand. Mainly the review articles helped identify types of machine learning algorithms for forecasting electricity demand. These models consider only historical electricity demand data and other factors such as population size, economic structure, season, month, day, and time of day. Therefore, no special attention has been paid to past weather data for forecasting electricity demand. Also, electricity demand hasn't been forecasted regionally using weather data in the Sri Lankan context.

B. The Impact of Weather Patterns on Electricity Demand

The main objective of this study is to forecast electricity demand in Sri Lanka using weather patterns through a machine-learning approach. Many studies have been conducted internationally on electricity demand, or electricity consumption or electricity load forecasting using the influence of the weather. Such studies have raised the importance of considering the impact of weather patterns on electricity demand data. The study has proposed an Autoregressive distributed lag(ARDL) model incorporating weather patterns to forecast monthly electricity consumption in Sri Lanka. This study has confirmed the importance of awareness of weather impacts on electricity demand [1]. Another study has emphasized that electricity demand increases when the temperature drops below 10 degrees Celsius due to a home's heating needs. Also, when the temperature rises above 23 degrees Celsius, the electricity demand also increases due to cooling requirements. This paper has explored the importance of using historical weather data (temperature, humidity) to forecast electricity loading in Sydney and New South Wales, Australia [24]. The annual temperature and the electricity demand have a significant positive correlation. While average rainfall is unrelated to electricity demand, relative humidity (%) is somewhat negatively connected with it[8]. Another study was done to forecast growing electricity demand and analyzed weather impacts on electricity demand in Saudi Arabia until 2040, using a weather-based long-term electricity demand forecasting model following a hybrid approach using end-use and econometric methods. The significance of climate

change awareness for upcoming purchases of heating and cooling equipment has been highlighted by this study [25]. It has been pointed out a temperature rise causes an increase in electric power consumption in Thailand[14]. A comprehensive analysis on an annual and seasonal basis reveals that meteorological parameters especially air temperature and dew point temperature have clearly influenced the electricity consumption in Muzaffarabad, Pakistan [26]. According to the above studies, a clear relationship with weather parameters can be seen in electricity demand. This study will consider electricity demand data as well as past weather data to achieve significant forecasting accuracy.

C. The specificity of machine learning approaches for accurate electricity demand forecasting using weather patterns

Many studies on electricity demand forecasting have been conducted internationally using different techniques. Several researchers have shown the importance of machine learning methods for forecasting electricity demand. The study used several time scales, and the findings revealed that by combining publicly accessible data with machine learning methods like linear regression and decision trees, it is possible to determine the electricity demand with high accuracy [27]. A review that was conducted to identify the type of model with the highest accuracy in the context of power forecasting. Mainly classical statistical or mathematical (MSC) (Autoregressive Integrated Moving Average, Classical Regression in the Time Series, Classical Statistical Model) and machine learning models, ANN, Recurrent Neural Networks, Fuzzy Neural Network-Based Forecasting Methods, Support Vector Machines models are compared on their Mean Absolute Percentage Error(MAPE) values. Among a large number of techniques, ANN is mostly used in the Machine Learning(ML) field, while the use of ARIMA models and regression analysis is prominent in the MSC area. According to the main findings, they concluded that ML models have better forecasting accuracy than traditional forecasting models [9]. ANN model has been compared with one of the basic time series SARIMA models. Based on the comparison, it was concluded that the ANN model is more suitable for forecasting the monthly electricity demand in Turkey using the effects of seasonal trends [28]. From the above evidence, it can be concluded that ML models often provide more accurate results for demand forecasting than other forecasting models.

III. METHODOLOGY

Various researchers have conducted experiments using different techniques to forecast electricity demand. This research's novelty is considering the correlation of weather variables in demand forecasting. The univariate case is expanded in the multivariate time series model by including two or more variables. It takes into account past data from other variables as well, not just its own. When multiple connected time series are observed simultaneously rather than just one, multivariate processes are created[29]. For that reason, multivariate time series models were considered in this study. According to the past, the majority of studies Artificial Neural Networks Model has mainly developed and given more accurate results related to forecasting electricity demand studies, [7]-[8] , [14]- [15]. Therefore, a multivariant LSTM model is developed for electricity demand forecasting, which is a widely used RNN architecture in time series forecasting.

The study has mentioned using a multivariate multistep LSTM model may obtain better prediction results [30]. In addition, the VAR model was developed to compare the forecasting performance of the LSTM model. The best-fitting model out of the two was selected based on the evaluation metrics. The methods followed in constructing the multivariate LSTM and VAR models are described below. Fig. 1 describes the proposed methodology of this study.

Models were developed in Jupyter Notebook with an Anaconda environment Python language was used as there are many open-source python libraries that can be useful for data manipulation, data visualization, statistics, mathematics, and machine learning. Python 3.9.13 version was used here.

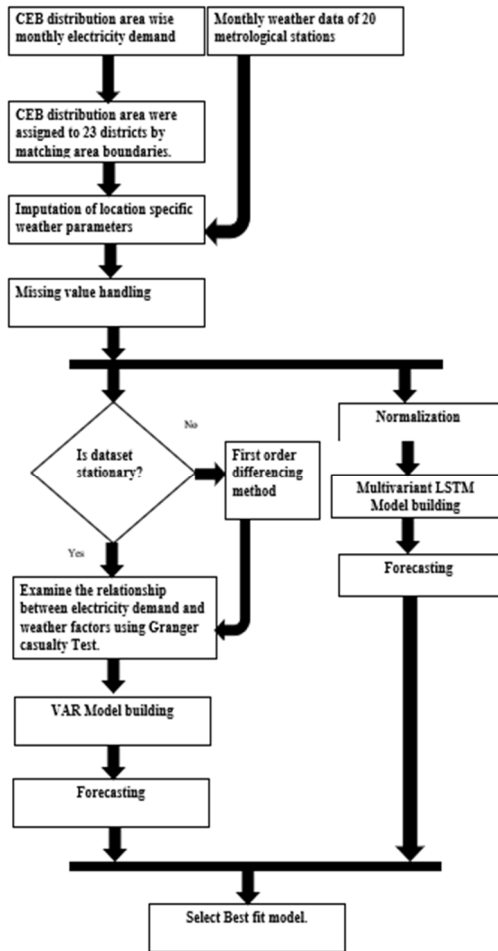


Fig. 1. Proposed methodology

A. Data Gathering

Monthly historical data of electricity demand in Sri Lanka and monthly weather data are used for the study. Electricity demand data was collected from the Statistics Department (of the Ceylon Electricity Board from January 2015 to December 2021). The CEB has declared 62 distribution areas covering the entire country, representing all these areas under four distribution divisions. Minimum and maximum air temperature, average temperature, humidity, and rainfall data are used as weather variables affecting electricity demand on a monthly basis. The data has been obtained from the Department of Metrology along with their latitude and longitude information representing 20 meteorological stations across the country. These 62 CEB areas were assigned to 23

districts by matching the CEB area boundaries with the district boundaries. But the remaining two districts of the Northern Province were ignored because there are no corresponding CEB areas. Then, the corresponding values of weather factors were assigned with respect to the centroid of each district. Then 23 districts were used to forecast the monthly electricity demand.

B. Data Pre-processing

1) Missing Value Handling

According to the weather data set that was received, there are more missing values in some districts weather data set. Therefore, SimpleImputer was used, a class in the scikit-learn library that imputes missing values in a dataset. The mean imputation strategy was used to replace missing values with the mean of the available values in the corresponding feature.

2) Detecting and treating the outliers.

The dataset may contain outliers. That means the set may have more unusual values than normal values. It may be larger or smaller than normal values. This issue can be generated due to experimental error/human error, so, if necessary, steps are not used to treat outliers it will lead to the generation of the wrong mean, or median of the quantitative data [31]. Here, Box plot graphical technique was used to identify outliers in a data set. A box plot shows the distribution of a data set by displaying the median, quartiles, and outliers in a box-and-whisker format. Some districts saw some outliers in weather data and electricity demand data. So, after investigation, some outliers had to be kept but some were removed.

3) Normalization

Neural networks require scaling the input vector and normalizing the output vector so that the data is in the range [0–1]. Data normalization avoids high dispersion between errors and weights during the training process [16]. Here, the StandardScaler technique was used to scale the data during LSTM model development so that outliers do not affect it. Because some outlier values had to be kept as it is.

C. Model Building and Training

1) Multivariate LSTM Model

Multivariate LSTM is a type of LSTM model designed to handle time series data with multiple input variables or features. In contrast to univariate LSTM, which takes only one-time series sequences as input, multivariate LSTM takes several correlated time series sequences as input [32]. The main idea behind a multivariate LSTM model is to use the relationships between the different variables to improve the model's performance. By analyzing the relationships between the different variables, the model can learn to make more accurate predictions about the future values of the variables. A typical multivariate LSTM model involves several layers of LSTM cells, which are used to process the input sequences of data. The output from the LSTM layers is then fed into one or more fully connected layers, which are used to make the final predictions. In this case, electricity demand forecasting based on past demand takes as input not only past monthly demand data measurements but also other variables such as average monthly temperature, average monthly rainfall, and monthly humidity in multivariate LSTM. The model was trained using monthly demand, average monthly temperature, average

monthly rainfall, and monthly humidity as input variables and demand values as the target variables.

2) VAR Model

It is a statistical model that analyzes the relationship between two or more time series variables. According to this study, electricity demand is forecasted using the VAR model using its past values as well as past values of meteorological variables. The advantage of the VAR model is that it can capture the dynamic relationships between variables. Used to predict how changes in one variable will affect other variables over time [33]. This was a useful tool to predict and understand the interaction between electricity demand and other weather variables. Checking whether time series data is stationary or not is a critical step when using VAR models. In a VAR model, the assumption of stationarity is necessary to accurately estimate model parameters and make reliable forecasts. A stationary time series means that statistical properties such as mean, variance, and autocorrelation structure remain constant over time. Non-stationary means those properties change over time, such as a trend or seasonality. In the development of this model in my study, Augment Dicky-fuller (ADF) test was used to check whether the weather data set is stationary or not. The null hypothesis of the ADF test is that the time series has a unit root, which means it is non-stationary. If the null hypothesis is rejected, it means that the time series is stationary. In ADF test p value is tested. P value means it measures the probability of observing the test statistic if the null hypothesis is true. If the p-value is less than the significance level (usually 0.05), the null hypothesis can be rejected, and the time series is considered stationary. If the p-value is greater than the significance level, the null hypothesis cannot be rejected, and the time series is considered non-stationary [34]. According to this study, every district's data set was not stationary over time. Therefore, the datasets had to be stationary. The data set was transformed to stationary using the first-order differencing method. When applying the first-order differencing method, the function subtracts each time series value from the previous value. This creates a new set of differences between successive values of the original series [35]. The next milestone in the VAR model is the Granger Causality test. It is a statistical concept that measures the predictive ability of one time series over another [36]. The test captures the dynamic relationships between the variables, thereby determining whether weather variable series are useful in predicting electricity demand and whether there is a causal relationship between the two. After finding the most correlated weather variables, the model was trained using those selected variables and demand values as input variables.

D. Model Validation and Evaluation

Model validation is the process of confirming that the model achieves its intended goal. Here collected data set should be mainly divided into two parts called the training set and the test set. The training dataset is used to train the proposed model, and the testing set measures the performance. The LSTM model was used 80% for training purposes and 20% for testing the model. In the VAR model, the last 12 values of the dataset were used for testing, and the remaining values were used for training. After model fitting, evaluation metrics were used to assess the accuracy of these two models. They are, mean square error (MSE), root mean square error

(RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE).

IV. RESULTS AND DISCUSSION

The main point of this research is to forecast monthly electricity demand by considering monthly weather factors in districts. The one of the main milestone in that objective is to find the correlation between weather variables and electricity demand.

A. Examination of the relationship between electricity demand and weather variables

The Granger causality test was used in the VAR model-building process to find out whether the weather variable time series is useful in predicting the electricity demand time series. After selecting a specific district, it is first examined which weather variables are responsible for predicting electricity demand in that district. For that, weather factors such as average temperature, minimum temperature, maximum temperature, average rainfall and day and night humidity were considered.

The p-value of the Granger causality test measures the statistical significance of the test results. It represents the probability of obtaining the observed test statistic if the null hypothesis (no Granger causality) is true. The p-value indicates the strength of the evidence against the null hypothesis. A low p-value (usually less than 0.05) suggests strong evidence to reject the null hypothesis and concludes that Granger causality exists between the two-time series. According to the above figure, an idea can be obtained that the average, minimum, and maximum temperatures have obtained a p-value close to 0.05. It means that the temperature and weather factor depends on the electricity demand. As other weather variables (rainfall, day and night humidity) have significantly large p values, it can be concluded that there is no dependence on those weather variables to determine the value of electricity demand regarding to Anuradhapura district. Similarly, all the p values obtained for all the districts are shown in Fig. 2 below.

According to the Fig.2 table, some humidity night columns are not filled because the missing values are more in that district humidity night column. Therefore, that column has been dropped for that district. A careful analysis of the above table shows that the average, minimum, and maximum temperatures have low p-values. Most of the p-values were less than 0.05. The northern province districts got a meager p-value for temperature than others. There is a strong dependency on the temperature in deciding the electricity demand value in that area. Other than that, Nuwara Eliya, and Kandy, like some districts, obtained a low p-value for nighttime humidity. It means the night humidity somewhat depends on those districts' electricity demand value. But in all cases, the rainfall weather factor obtained higher p-values. It implies that the rainfall weather variable does not affect the determination of electricity demand value in Sri Lanka. Finally, it can say humidity also has some minor effects on the demand figure, but it varies from district to district. But the rainfall weather factor is not affected the electricity demand in Sri Lanka. But the temperature is the dominant weather factor affecting the electricity demand in Sri Lanka.

District	Average Temperature	Maximum Temperature	Minimum Temperature	Rainfall	Day Humidity	Night Humidity
Anuradhapura	0.706	0.0857	0.791	0.762	0.3772	0.5067
Polonnaruwa	0.0278	0.0284	0.04	0.217	0.2637	-
Colombo	0.2031	0.0208	0.173	0.2586	0.392	0.2126
Gampaha	0.0749	0.0179	0.0855	0.4318	0.5831	0.5833
Kalutara	0.187	0.0531	0.0837	0.2746	0.1975	0.3515
Galle	0.0501	0.1017	0.0139	0.3922	0.0411	0.092
Matara	0.5201	0.5312	0.313	0.5274	0.1354	0.028
Hambantota	0.2959	0.0647	0.7244	0.1666	0.2388	0.2669
Rathnapura	0.0267	0.0198	0.0379	0.1296	0.0245	0.2873
Kegalle	0.0966	0.2761	0.0548	0.0505	0.0878	-
Badulla	0.0654	0.2011	0.0602	0.2371	0.2352	-
Monaragala	0.129	0.0171	0.0363	0.1736	0.209	
Puttalam	0.282	0.2831	0.1206	0.6072	0.3286	0.2487
Kurunegala	0.1387	0.1414	0.0991	0.1771	0.2222	-
Nuwara Eliya	0.0414	0.0477	0.0462	0.1629	0.3424	0.0746
Kandy	0.0452	0.2194	0.0276	0.2887	0.3624	0.0267
Matale	0.0411	0.1705	0.0299	0.6015	0.6229	0.3277
Ampara	0.0855	0.0855	0.0855	0.4061	0.3219	-
Batticaloa	0.0014	0.0038	0.0013	0.6176	0.2182	0.747
Trincomalee	0.004	0.0096	0.0068	0.2994	0.2454	0.8679
Jaffna	0.0001	0.0001	0.0013	0.3141	0.1654	-
Vavuniya	0.0123	0.0146	0.0167	0.4124	0.2545	-
Kilinochchi	0.0177	0.12	0.431	0.7221	0.4578	0.4433

Fig. 2 . District wise p-values

B. Results of Machine Learning Models

After VAR, and LSTM model training, the following results were obtained for the test data set related to the Anuradhapura districts. Fig.3 shows the prediction results related to the test data set related to Anuradhapura districts related to VAR model. Fig.4 shows the prediction results related to the test data set related to Anuradhapura districts as a graph representation.

	Demand	Temp_Avg	Demand_predicted	Temp_Avg_predicted
2021-01-01	19.324695	26.3	17.959786	27.090562
2021-02-01	18.543998	26.9	17.352181	27.927133
2021-03-01	20.638740	28.4	18.025990	29.024670
2021-04-01	22.340633	29.5	19.389260	29.432383
2021-05-01	21.584340	28.9	19.903125	29.341620
2021-06-01	20.875111	29.1	20.866339	28.934976
2021-07-01	22.094731	29.1	20.965888	28.417947
2021-08-01	20.739739	29.0	19.848377	28.482735
2021-09-01	21.633316	29.2	19.523451	28.544384
2021-10-01	21.214337	27.9	19.900186	28.458630
2021-11-01	20.385244	26.9	20.384627	27.838049
2021-12-01	20.907689	26.7	19.842807	27.250335

Fig. 3. Predicted demand values of VAR model.

Fig.5 shows the prediction results related to the test data set related to Anuradhapura districts related to LSTM model.

Fig.6 shows the prediction results related to the test data set as a graph representation.

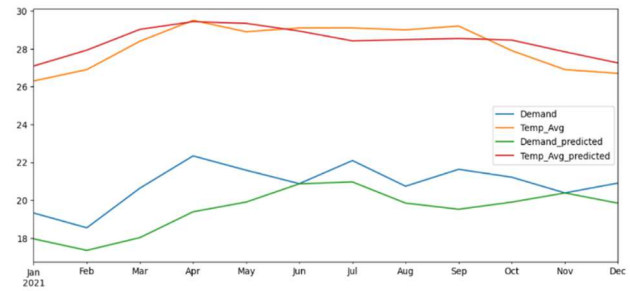


Fig. 4. The visualization of predicted demand values of VAR model

For example, the below Fig.7 shows the comparison of evaluation metrics related to the Anuradhapura district. According to this table, the VAR model gives a lower value than the LSTM model for all evaluation metrics. Also, in other districts, the VAR model is given a lower value than the LSTM model. Based on that, it can be concluded that the VAR model is better than LSTM model for electricity demand

forecasting using weather variables.

	Demand	Temp_Avg	Rainfall_Avg	HumidityDay	HumidityNight	Demand_Predicted
Month						
2021-02-01	18.543998	26.9	19.8	69	92.0	19.113526
2021-03-01	20.638740	28.4	40.5	67	91.0	19.008534
2021-04-01	22.340633	29.5	131.0	67	91.0	18.994457
2021-05-01	21.584340	28.9	90.6	74	92.0	19.121771
2021-06-01	20.875111	29.1	3.0	68	89.0	19.503055
2021-07-01	22.094731	29.1	72.1	68	88.0	19.650974
2021-08-01	20.739739	29.0	106.2	66	87.0	19.842489
2021-09-01	21.633316	29.2	132.6	67	89.0	19.707705
2021-10-01	21.214337	27.9	276.5	77	93.0	19.544722
2021-11-01	20.385244	26.9	413.6	82	95.0	19.409553
2021-12-01	20.907689	26.7	54.0	76	94.0	18.928156

Fig. 5. Predicted demand values of LSTM model.

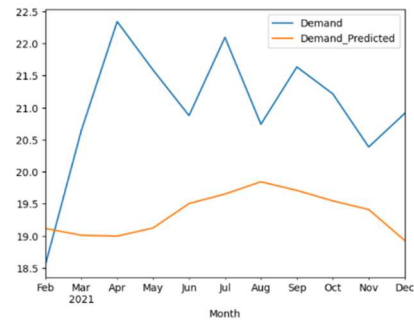


Fig. 6. The visualization of predicted demand values of LSTM model

	MSE	RMSE	MAE	MAPE
LSTM	3.6607	1.9133	1.7519	0.0821
VAR	1.4999	1.2245	0.9725	0.0429

Fig. 7. Comparison of models

V. CONCLUSION & FUTURE WORK

The present study has found the best time series model to forecast monthly electricity demand in Sri Lanka, considering weather patterns. Along the way, this study explored the impact of weather-related factors on electricity demand. Based on that, it was determined that temperature is the main weather factor that determines electricity demand in Sri Lanka. Among the multivariate LSTM and VAR models compared, the VAR model with weather-related forecasts is the most suitable for forecasting monthly electricity demand in Sri Lanka. The proposed VAR model predicts district-wise electricity demand, which helps in planning the electricity supply regionally. Therefore, we propose to use the VAR model for CEB regional electricity demand management.

In this research, only multivariate LSTM and VAR were used. However, many other multiple techniques can be used to forecast electricity demand. Examining and monitoring such practices will help create more accurate and precise forecasting mechanisms for electricity demand. In model development, monthly electricity demand and monthly weather variables are only considered input variables in this study. Only weather variables have been used as exogenous variables for this study. However, it would be interesting to analyze other variables, such as econometric variables (GDP, population), and see how they affect the electricity demand forecast. Then, many correlated variables can be identified by necessary feature engineering methods. Using weather variables along with other dependent variables is useful in providing a more accurate forecast of electricity demand.

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REFERENCES

- [1] A. D. A. D. L. R. S. K. D. & L. E. Y. K. Priyadarshana, "Using weather patterns to forecast electricity consumption in Sri Lanka: An ardl approach," 2021.
- [2] G. V. & S. L. De Silva, "Peak electricity demand prediction model for Sri Lanka power system," *Engineer: Journal of the Institution of Engineers, Sri Lanka*, p. 46(4), 2013.
- [3] T. a. G. P. B. E. B. S. L. The Generation Planning Unit, "CEB|Publications & Media," [Online]. Available: https://ceb.lk/front_img/img_reports/1636539187LTGEP_2022-2041_Web_compressed.pdf. [Accessed 4 September 2022].
- [4] M. & R. L. Danthurebandara, "Environmental consequences of different electricity generation mixes in Sri Lanka by 2050," *Journal of Cleaner Production*, pp. 210, 432-444, 2019.
- [5] "CEB|Business With CEB," [Online]. Available: <https://ceb.lk/electricity-generated/en>. [Accessed 14 December 2022].
- [6] S. e. a. Perera, "Supply and demand planning of electricity power: a comprehensive solution," *IEEE Conference on Information and Communication Technology*. IEEE, 2019.
- [7] S. L. & N. H. R. Karunatilake, "Artificial neural networks for daily electricity demand prediction of Sri Lanka," in *In 2017 Seventeenth International Conference on Advances in ICT for Emerging Regions (ICTer)*, 2017, September.
- [8] D. C. H. K. T. M. U. & J. A. G. B. P. Hapuarachchi, "Long term annual electricity demand forecasting in Sri Lanka by artificial neural networks," in *In 2018 IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC)*, 2018, October.
- [9] E. A.-C. H. & S. R. Vivas, "A systematic review of statistical and machine learning methods for electrical power forecasting with reported mape score," *Entropy*, pp. 22(12), 1412.
- [10] Z. C. F. Y. S. L. & Z. K. L. Shao, "A review of the decomposition methodology for extracting and identifying the fluctuation characteristics in electricity demand forecasting," *Renewable and Sustainable Energy Reviews*, pp. 75, 123-136, 2017.
- [11] N. S. M. S. A. & J. B. N. Salleh, "A systematic literature review of machine learning methods for short-term electricity forecasting," in *In 2020 8th International conference on information technology and multimedia (ICIMU)*, 2020, August.
- [12] M. S. D. R. C. A. J. F. & L. Y. Al-Musaylh, "Short-term electricity demand forecasting with MARS, SVR and ARIMA models using aggregated demand data in Queensland, Australia," *Advanced Engineering Informatics*, pp. 35, 1-16, 2018.
- [13] N. G. I. S. H. K. T. M. U. & J. A. G. B. P. Samarawickrama, "Support Vector Machine Regression for forecasting electricity demand for large commercial buildings by using kernel parameter and storage effect," in *In 2016 Moratuwa Engineering Research Conference (MERCon)*, 2016, April.
- [14] K. P. C. & K. D. Panklib, "Electricity consumption forecasting in Thailand using an artificial neural network and multiple linear regression," *Energy Sources, Part B: Economics, Planning, and Policy*, pp. 10(4), 427-434, 2015.
- [15] K. B. S. S. & S. P. Sahay, "Short-term load forecasting of Toronto Canada by using different ANN algorithms," in *In 2016 IEEE 6th International Conference on Power Systems (ICPS)*, 2016, March.
- [16] R. Z. M. & B.-S. O. Houimli, "Short-term electric load forecasting in Tunisia using artificial neural networks," 2020, pp. 11(2), 357-375, *Energy Systems*.
- [17] A. Manowska, "Using the LSTM network to forecast the demand for electricity in Poland," *Applied Sciences*, pp. 10(23), 8455, 2020.
- [18] H. S. M. & Y. M. Abbasimehr, "An optimized model using LSTM network for demand forecasting," *Computers & industrial engineering*, pp. 143, 106435, 2020.
- [19] Y. R. M. C. & S. R. Liu, "A vector autoregression weather model for electricity supply and demand modeling," *Journal of Modern Power Systems and Clean Energy*, pp. 6(4), 763-776, 2018.
- [20] S. S. e. a. Jones, "A multivariate time series approach to modeling and forecasting demand in the emergency department," *Journal of biomedical informatics* 42.1 , pp. 123-139, 2009.
- [21] A. & E. H. A. Salam, "Comparison of machine learning algorithms for the power consumption prediction:-case study of tetouan city-," in *In 2018 6th International Renewable and Sustainable Energy Conference (IRSEC)*, 2018, December.
- [22] C. S. I. F. I. & S. G. Nichiforov, "Energy consumption forecasting using ARIMA and neural network models," in *In 2017 5th International Symposium on Electrical and Electronics Engineering (ISEEE)*, 2017, October.
- [23] K. & K. A. B. Goswami, "Electricity demand prediction using data driven forecasting scheme: ARIMA and SARIMA for real-time load data of Assam," in *In 2020 International Conference on Computational Performance Evaluation (ComPE)* , 2020, July.
- [24] V. K. A. K. M. L. & Z. A. Dehalwar, "Electricity load forecasting for Urban area using weather forecast information," in *In 2016 IEEE International Conference on Power and Renewable Energy (ICPRE)*, 2016, October.
- [25] N. a. J. N. Alabbas, "Weather-Based Long-Term Electricity Demand Forecasting Model for Saudi Arabia: A Hybrid Approach Using End-Use and Econometric Methods for Comprehensive Demand Analysis," in *34th USAEE/IAEE North American Conference*, 2016.
- [26] M. N. M. S. A. S. S. O. K. I. R. S. A. H. N. .. & A. W. Jawad, "Machine learning based cost effective electricity load forecasting model using correlated meteorological parameters," *IEEE Access*, pp. 8, 146847-146864, 2020.
- [27] Z. & G. M. C. Camurdan, "Machine learning based electricity demand forecasting," in *In 2017 International Conference on Computer Science and Engineering (UBMK)*, 2017, October.
- [28] C. E. H. A. & Ç. R. Hamzaçebi, "Forecasting of Turkey's monthly electricity demand by seasonal artificial neural network. *Neural Computing and Applications*," pp. 31, 2217-2231, 2019.

- [29] I. A. & O. A. S. Iwok, "A comparative study between univariate and multivariate linear stationary time series models," *American Journal of Mathematics and Statistics*, pp. 6(5), 203-212, 2016.
- [30] E. Z. H. G. X. Z. L. Z. & L. Y. Yang, "A multivariate multi-step LSTM forecasting model for tuberculosis incidence with model explanation in Liaoning Province, China.," *BMC Infectious Diseases*, pp. 22(1), 490, 2022.
- [31] "7.1.6. What are outliers in the data?," [Online]. Available: <https://www.itl.nist.gov/div898/handbook/prc/section1/prc16.htm>. [Accessed 24 February 2023].
- [32] "Multi-variate Time Series Forecasting using Kats Model," [Online]. Available: <https://www.analyticsvidhya.com/blog/2022/08/multi-variate-timeseries-forecasting-using-kats-model/>. [Accessed 23 February 2023].
- [33] "Introduction to the Fundamentals of Vector Autoregressive Models - Aptech," [Online]. Available: <https://www.aptech.com/blog/introduction-to-the-fundamentals-of-vector-autoregressive-models/>. [Accessed 23 February 2023].
- [34] "Augmented Dickey Fuller Test (ADF Test) – Must Read Guide - ML+," [Online]. Available: <https://www.machinelearningplus.com/time-series/augmented-dickey-fuller-test/>. [Accessed 23 February 2023].
- [35] "8.1 Stationarity and differencing |Forecasting:Principles and practice (2nd ed)," [Online]. Available: <https://otexts.com/fpp2/stationarity.html>. [Accessed 23 February 2023].
- [36] "A Quick Introduction On Granger Causality Testing For Time Series Analysis|by Susan Li |Towards Data Science," [Online]. Available: <https://towardsdatascience.com/a-quick-introduction-on-granger-causality-testing-for-time-series-analysis-7113dc9420d2>. [Accessed 20 February 2023].
- [37] C. e. a. Nichiforov, "Energy consumption forecasting using ARIMA and neural network models.," in 2017 5th International Symposium on Electrical and Electronics Engineering (ISEEE). IEEE, 2017.
- [38] V. e. a. Dehalwar, "Electricity load forecasting for Urban area using weather forecast information," in IEEE International Conference on Power and Renewable Energy (ICPRE). IEEE, 2016.